

ExpenseTracker

Project Report — Flask-Based Personal Expense Tracker

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Technology Stack	Python, Flask, SQLAlchemy, SQLite, Bootstrap, Chart.js

1. Project Overview

ExpenseTracker is a Flask web application that lets a user log personal expenses, set a monthly budget, and track spending by category. It demonstrates a full CRUD workflow, user authentication, and data visualization within a single-user web application architecture.

2. Key Features

- User registration and login (Flask-Login, hashed passwords via Werkzeug)
- Add, edit, and delete expenses with title, amount, category, date, and note
- Monthly budget entry with a live progress bar and remaining-balance calculation
- Dashboard summarizing the current month's total spend and category-wise breakdown
- Category-based filtering on the full expense list
- Analytics page with monthly spend, yearly spend, and all-time category breakdown bar charts
- ML-based spend forecasting (moving average + linear regression) with confidence scoring
- JSON REST API endpoint (/api/expenses) for programmatic access

3. Technology Stack

Layer	Technology
Backend	Flask (Python)
Database / ORM	SQLite + Flask-SQLAlchemy
Authentication	Flask-Login, Werkzeug password hashing
Frontend	Jinja2 templates, Bootstrap 5 (self-hosted, no external CDN)
Data Visualization	Chart.js — monthly/yearly/category bar charts, forecast line chart
ML Forecasting	scikit-learn LinearRegression + moving-average baseline

4. Database Design

User — id, email, password_hash, monthly_budget

Expense — id, user_id (FK), title, amount, category, note, expense_date, created_at

5. Spend Forecasting Methodology

Next month's spend is estimated using two comparative methods, shown side by side rather than picking one: a **moving average** (mean of the last up to 3 months — a stable baseline that lags behind a real trend) and a **linear regression** (scikit-learn) fitted on monthly totals against a month index, which captures the trend but is more sensitive with short history. A confidence label (low / medium / high) is shown based on how many months of history are available, so the prediction does not look falsely precise when there is little data to support it.

6. Application Screenshots

The following screenshots were captured from the actual running application (via an automated browser) with sample seeded data spanning 12 months.

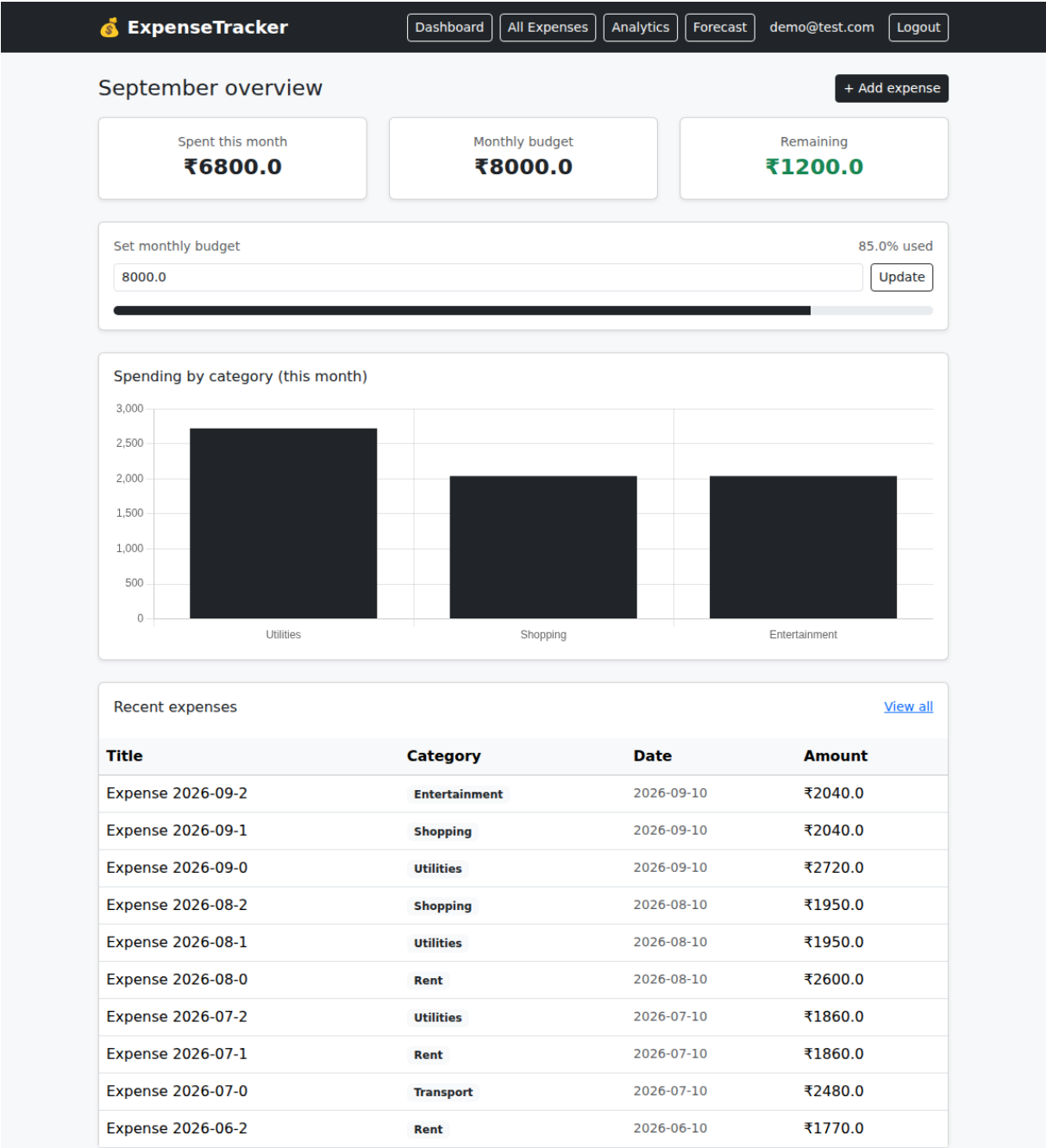


Figure 1: Dashboard — monthly overview, budget progress, category chart, recent expenses

Analytics

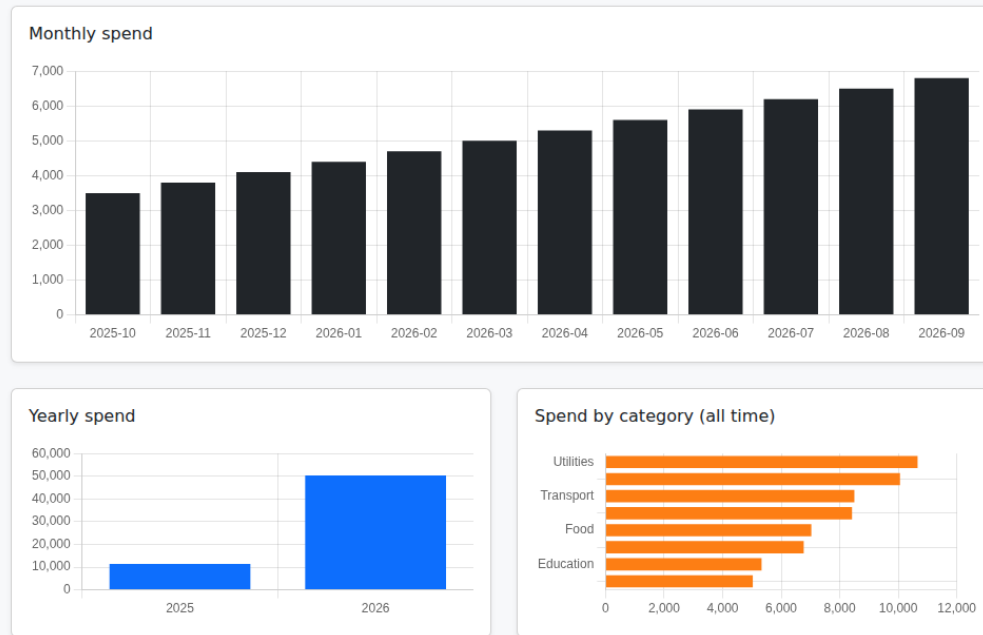


Figure 2: Analytics — monthly spend, yearly spend, and category breakdown bar charts

Spend forecast

Moving average forecast

₹6500.0

Average of last 3 month(s)

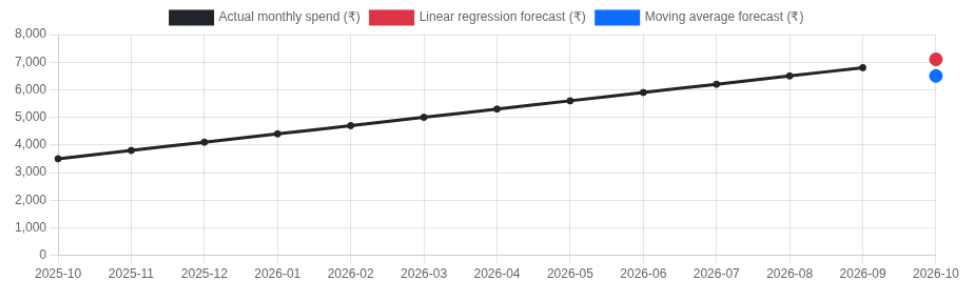
Linear regression forecast

₹7100.0

Trend: +₹300.0/month

Confidence: High. Based on 12 months of data — a reasonably grounded estimate.
Forecast is for **2026-10**.

Monthly spend history



Month	Total spend
2025-10	₹3500.0
2025-11	₹3800.0
2025-12	₹4100.0
2026-01	₹4400.0
2026-02	₹4700.0
2026-03	₹5000.0
2026-04	₹5300.0
2026-05	₹5600.0
2026-06	₹5900.0
2026-07	₹6200.0
2026-08	₹6500.0
2026-09	₹6800.0

Figure 3: Spend Forecast — moving average vs linear regression, with confidence level

ExpenseTracker

Dashboard

All Expenses

Analytics

Forecast

demo@test.com

Logout

All expenses (₹61800.0 total)

+ Add expense

All

Food

Transport

Rent

Utilities

Shopping

Entertainment

Health

Education

Other

Title	Category	Date	Note	Amount		
Expense 2026-09-2	Entertainment	2026-09-10	—	₹2040.0	Edit	Delete
Expense 2026-09-1	Shopping	2026-09-10	—	₹2040.0	Edit	Delete
Expense 2026-09-0	Utilities	2026-09-10	weekly log	₹2720.0	Edit	Delete
Expense 2026-08-2	Shopping	2026-08-10	—	₹1950.0	Edit	Delete
Expense 2026-08-1	Utilities	2026-08-10	—	₹1950.0	Edit	Delete
Expense 2026-08-0	Rent	2026-08-10	weekly log	₹2600.0	Edit	Delete
Expense 2026-07-2	Utilities	2026-07-10	—	₹1860.0	Edit	Delete
Expense 2026-07-1	Rent	2026-07-10	—	₹1860.0	Edit	Delete
Expense 2026-07-0	Transport	2026-07-10	weekly log	₹2480.0	Edit	Delete
Expense 2026-06-2	Rent	2026-06-10	—	₹1770.0	Edit	Delete
Expense 2026-06-1	Transport	2026-06-10	—	₹1770.0	Edit	Delete
Expense 2026-06-0	Food	2026-06-10	weekly log	₹2360.0	Edit	Delete
Expense 2026-05-2	Transport	2026-05-10	—	₹1680.0	Edit	Delete
Expense 2026-05-1	Food	2026-05-10	—	₹1680.0	Edit	Delete
Expense 2026-05-0	Education	2026-05-10	weekly log	₹2240.0	Edit	Delete
Expense 2026-04-2	Food	2026-04-10	—	₹1590.0	Edit	Delete
Expense 2026-04-1	Education	2026-04-10	—	₹1590.0	Edit	Delete
Expense 2026-04-0	Health	2026-04-10	weekly log	₹2120.0	Edit	Delete
Expense 2026-03-2	Education	2026-03-10	—	₹1500.0	Edit	Delete
Expense 2026-03-1	Health	2026-03-10	—	₹1500.0	Edit	Delete
Expense 2026-03-0	Entertainment	2026-03-10	weekly log	₹2000.0	Edit	Delete
Expense 2026-02-2	Health	2026-02-10	—	₹1410.0	Edit	Delete
Expense 2026-02-1	Entertainment	2026-02-10	—	₹1410.0	Edit	Delete
Expense 2026-02-0	Shopping	2026-02-10	weekly log	₹1880.0	Edit	Delete
Expense 2026-01-2	Entertainment	2026-01-10	—	₹1320.0	Edit	Delete
Expense 2026-01-1	Shopping	2026-01-10	—	₹1320.0	Edit	Delete
Expense 2026-01-0	Utilities	2026-01-10	weekly log	₹1760.0	Edit	Delete
Expense 2025-12-2	Shopping	2025-12-10	—	₹1230.0	Edit	Delete
Expense 2025-12-1	Utilities	2025-12-10	—	₹1230.0	Edit	Delete
Expense 2025-12-0	Rent	2025-12-10	weekly log	₹1640.0	Edit	Delete
Expense 2025-11-2	Utilities	2025-11-10	—	₹1140.0	Edit	Delete
Expense 2025-11-1	Rent	2025-11-10	—	₹1140.0	Edit	Delete
Expense 2025-11-0	Transport	2025-11-10	weekly log	₹1520.0	Edit	Delete
Expense 2025-10-2	Rent	2025-10-10	—	₹1050.0	Edit	Delete
Expense 2025-10-1	Transport	2025-10-10	—	₹1050.0	Edit	Delete
Expense 2025-10-0	Food	2025-10-10	weekly log	₹1400.0	Edit	Delete

Figure 4: All Expenses — full list with category filters

 ExpenseTracker

Dashboard

All Expenses

Analytics

Forecast

demo@test.com

Logout

Add expense

Title

e.g. Groceries

Amount (₹)

Category

Food

▼

Date

09/01/2026

Note (optional)

Add expense

Figure 5: Add Expense form